

Ben Cookson

PHD STUDENT · UNIVERSITY OF TORONTO

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Education

University of Toronto

PHD + MSc COMPUTER SCIENCE

- Advisors: Allan Borodin and Nisarg Shah

Toronto, Ontario, Canada

Sept 2023 - Present

University of New Brunswick

BACHELOR OF COMPUTER SCIENCE

- GPA: 4.3/4.3

Fredericton, NB, Canada

Sept 2018 - May 2023

Research Profile

Research interests: Social choice theory, fair division and resource allocation, algorithmic game theory, mechanism design, and online algorithms.

Affiliations: Schwartz Reisman Institute for Technology and Society (graduate affiliate; Graduate Fellow, 2025–2026) and Vector Institute.

Publications

Full versions of all papers can be found at bcookson.github.io

WORKING PAPERS

- W1. **Improved Lower Bounds for Proportionally Fair Clustering**, Benjamin Cookson, Eva Deltl, Yeeseok Oh, 2026. Working paper.

CONFERENCE PAPERS

- C1. **Temporal Fair Division**, Benjamin Cookson, Soroush Ebadian, Nisarg Shah, in Proc. of 39th Annual AAAI Conf. on Artificial Intelligence (AAAI'25), 2025. (Vol. 39, No. 13, pp. 13727–13734).
- C2. **Constrained Fair and Efficient Allocations**, Benjamin Cookson, Soroush Ebadian, Nisarg Shah, in Proc. of 39th Annual AAAI Conf. on Artificial Intelligence (AAAI'25), 2025. (Vol. 39, No. 13, pp. 13718–13726). *Oral presentation (top 4.6% of submissions)*.
- C3. **Fairly Stable Two-Sided Matching with Indifferences**, Benjamin Cookson, Nisarg Shah, in Proc. of Twenty-Sixth ACM Conference on Economics and Computation, 2025 (p. 1130).
- C4. **Unifying Proportional Fairness in Centroid and Non-Centroid Clustering**, Benjamin Cookson, Nisarg Shah, Ziqi Yu, in Proc. of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS'25), 2025. *Spotlight presentation (top 3.2% of submissions)*.
- C5. **Fairness Perceptions of Large Language Models**, Benjamin Cookson, Soroush Ebadian, Nisarg Shah, in Proc. of 40th Annual AAAI Conf. on Artificial Intelligence (AAAI'26), 2026.
- C6. **Proportional Fairness for Harmful Decisions**, Benjamin Cookson, Soroush Ebadian, Dominik Peters, Nisarg Shah, in Proc. of the 27th ACM Conference on Economics and Computation (EC'26), 2026.
- C7. **Constrained Fair Allocations via Partition Matroid Reductions**, Benjamin Cookson, Nisarg Shah, in Proc. of the 6th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO'26), 2026.

Professional Experience

2023-2025 **Research Assistant**, University of Toronto
2023-2025 **Graduate Teaching Assistant**, University of Toronto
2022-2023 **Cybersecurity Student Researcher**, National Research Council Canada
2020-2023 **Software Developer Intern**, IBM Canada

Awards, Fellowships, & Grants

2026	Canada Graduate Research Scholarship – Doctoral (CGRS D) , Natural Sciences and Engineering Research Council of Canada (NSERC)	<i>\$40,000/year (36 months)</i>
2025–2026	SRI Graduate Fellow , Schwartz Reisman Institute for Technology and Society	<i>\$7,500</i>
2023–2025	Ontario Graduate Scholarship , Government of Ontario	<i>\$30,000</i>
2023	Lieutenant-Governor of New Brunswick Silver Medal , University of New Brunswick	
2023	Govind and Lakshmi Gujar Computer Science Prize , University of New Brunswick	<i>\$1,000</i>

Teaching Experience

2024–2026 **CSC373 - Algorithm Design, Analysis and Complexity**, Teaching Assistant, University of Toronto
2024 **CSC240 - Enriched Introduction to the Theory of Computation**, Teaching Assistant, University of Toronto
2023 **CSC110 - Foundations of Computer Science I**, Teaching Assistant, University of Toronto

Professional Service

Journal Reviewer For: ACM Transactions on Economics and Computation (TEAC), Games and Economic Behavior (GEB), Algorithmica, Artificial Intelligence (AIJ)

Conference Reviewer For: EC 2026, AAAI 2026, FAccT 2025